

STAREHE BOYS' 2025 PREDICTION



Kenya Certificate of Secondary Education

ANTICIPATED KCSE EXAM



233/3

Chemistry (Practical)

Paper 3

Time: 2¼ Hours

SCHOOL..... SIGN.....

CONFIDENTIAL

1. Apart from the laboratory fittings supply the following

- (i). 100cm³ solution A - 0.02M Acidified potassium manganite (VII) ($H^+/KMnO_4$)
- (ii). 100cm³ Solution B - 0.05M Oxalic acid
- (iii). 100cm³ Solution C - containing 4.9g of $(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$ in 250cm³ of solution)
- (iv). 50ml measuring cylinder
- (v). 5 Test tubes
- (vi). 1 boiling tube
- (vii). 250ml glass beaker
- (viii). Thermometer
- (ix). Stopwatch
- (x). Burette
- (xi). Pipette
- (xii). Two conical flasks
- (xiii). About 1g of solid Q ($ZnSO_4$)
- (xiv). Distilled water
- (xv). About 0.5g of solid L (oxalic acid)

(xvi). *Litmus papers*

(xvii). *Metallic spatula*

(xviii) *Means of Heating*

(xix) *1g sodium carbonate*

Access to

- i.** *1 M Sodium hydroxide solution*
- ii.** *1M Ammonia solution*
- iii.** *0.5M Barium Nitrate solution – Acidified*
- iv.** *1M sulphuric (VI) acid*
- v.** *Acidified potassium Manganate (VII)*

NB

0.02M of H^+/KMnO_4 is prepared by dissolving
3.16g of KMnO_4 in 400cm^3 of $2\text{M H}_2\text{SO}_4$.
Add distilled water to make 1 litre of
solution.

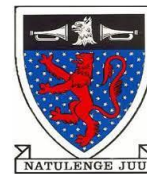


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Time: 2¼ Hours

NAME.....

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INDEX NO..... ADM NO.....

INSTRUCTIONS:

- Answer **ALL** questions in the spaces provided.
- You are **NOT** allowed to start working with the apparatus for the first 15 minutes of the 2 ¼
- Hours allowed for this paper. This time will enable you read through the question paper and make sure you have all the chemicals and apparatus required.
- Mathematical tables and electronic calculators may be used
- All working must be clearly shown where necessary.

Answer **ALL** questions in the spaces provided.

1. You are provided with:

- Solution A of Potassium manganate (VII).
- 0.05M solution B of oxalic acid.
- Solution C containing 4.9g of ammonium iron (II) Sulphate, $(\text{NH}_4)_2 \text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$, in 250cm^3 of water.
- You are required to:
 - i) Determine the rate of reaction between oxalic acid and Potassium manganate (VII).
 - ii) Standardize the solution A.

PROCEDURE I:

Using a measuring cylinder, place 1 cm^3 of solution A into each of the five (5) test-tubes in a rack.

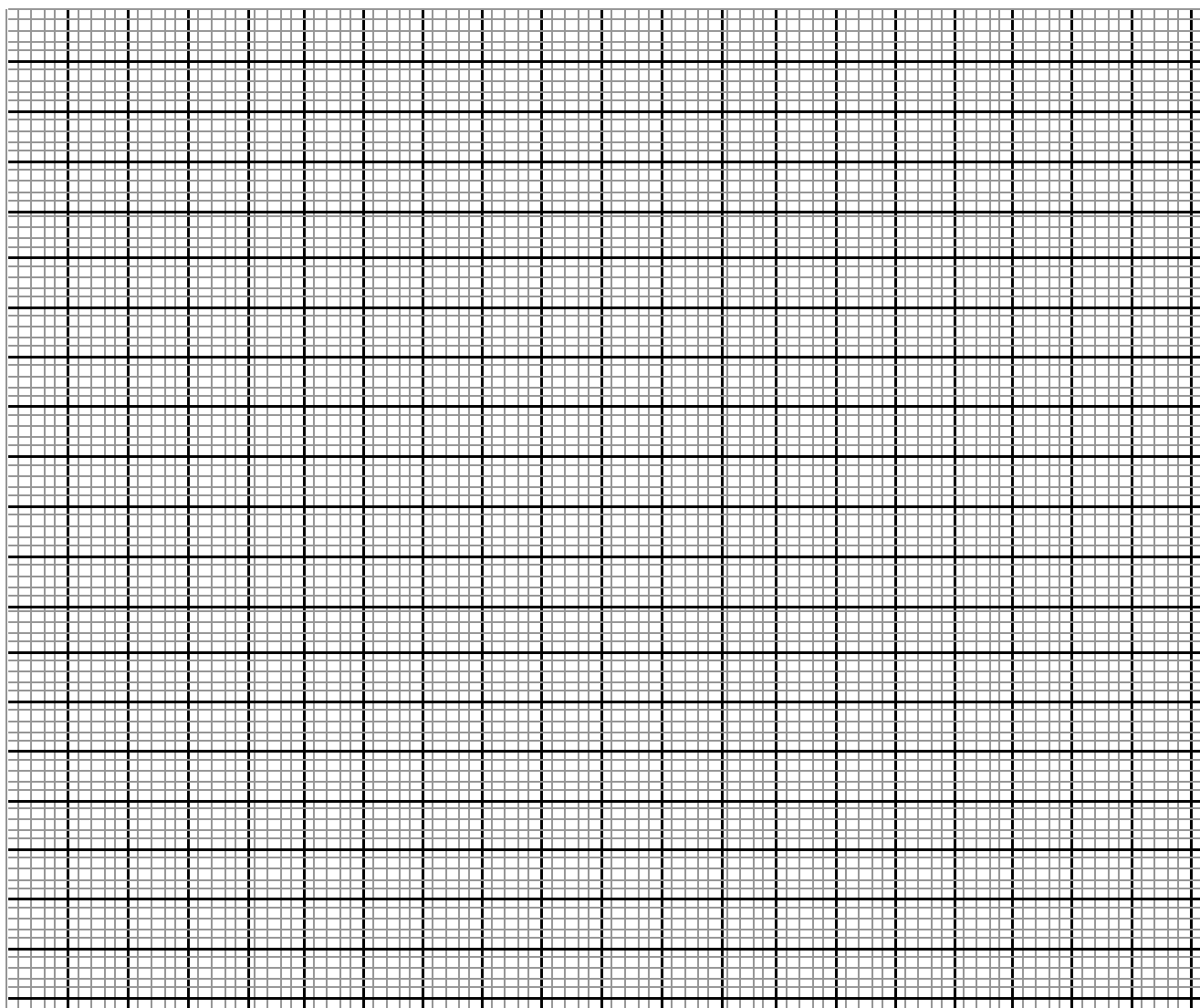
Clean the measuring cylinder and use it to place 19cm^3 of solution B into a boiling tube. Prepare a water bath by placing about 200cm^3 of water into a beaker and start to heat. Place a thermometer into solution B and place it in the warm water until it attains a temperature of 40°C . Remove the boiling tube from the water-bath and place it in the test-tube rack. Add the first portion of solution A immediately and at the same time start a stop watch. Record the time taken of solution A to be decolourised in table I below. Repeat the procedure at temperatures of 50°C , 60°C , 70°C and 80°C to complete the table.

Temperature of solution B ($^\circ\text{C}$)	40	50	60	70	80
Time taken for decolourisation (tsecs)					
$1/t \text{ sec}^{-1}$					

(3 marks)

i) Plot a graph of $1/t$ against temperature (X-axis)

(3 marks)



ii) From the graph determine the time taken for the mixture to decolourise at 65°C (2 marks)

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iii) How does the rate of reaction between oxalic acid and Potassium manganate (VII) vary with

temperature? (1 mark)

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PROCEDURE II

Fill a burette with solution A. Pipette 25cm³ of solution C into a conical flask and titrate the solution A against solution C until a permanent pink colour just appears. Record your results in table II below and repeat the procedure to fill the table.

	I	II	III
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of solution A used (cm ³)			

(4 marks)

i) Determine the average volume of A usedcm³

(Show how you arrive at your answer) (1 mark)

ii). Calculate the concentration of solution C in moles per litre (Fe, = 56, S = 32,

O = 16, N =14, H = 1) (2 marks)

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iii) Find the number of moles of solution C used.

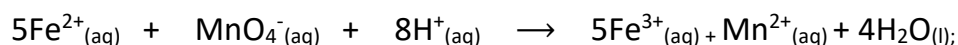
(1 mark)

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iv) Given the ionic equation for the reaction is



Find the number of moles of solution A used.

(1 mark)

(v) Determine the concentration of the Potassium manganate (VII), solution A in moles per litre.

(2 marks)

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2. You are provided with solid Q. Carry out the tests below and record your observations and inferences in the table below.

i) Place half a Spatula full of solid Q in a clean dry test-tube and heat gently then Strongly.

Observations	Inferences
(1 mark)	(1 mark)

- ii). Place the remaining solid Q in a boiling tube and add about 5cm³ of distilled water and shake well.
Divide the resulting mixture into four portions for the tests below.

Observations	Inferences
(1 mark)	(1 mark)

- a). To the first portion add Sodium hydroxide solution dropwise until in excess.

Observations	Inferences
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(1 mark)	(1 mark)
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b). To the second portion add 2-3 drops of dilute Sulphuric (VI) acid

Observations	Inferences
(1 mark)	(1 mark)

c). To the third portion add aqueous ammonia dropwise until in excess

Observations	Inferences
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(1 mark)	(1 mark)
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d). To the fourth portion add 2-3 drops acidified barium nitrate solution

Observations	Inferences
(1 mark)	(1 mark)

3. You are provided with solid L. carry out the tests below on L and record the observations and inferences in the spaces provided.

(a) Place half of solid L in a boiling tube and add about 5cm³ of distilled water

(i)

Observations	Inferences
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(1 mark)	(1 mark)
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Divide the solution into two portions and carry out the tests below.

(ii). To the first portion add 2-3 drops of acidified potassium manganate (VII).

Observations	Inferences
(1 mark)	(1 mark)

(iii). To the second portion add Sodium carbonate provided (1 mark)

Observations	Inferences

(1 mark)	(1 mark)
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(b). Place the remaining solid L in metallic spatula and ignite it.

Observations	Inferences
(1 mark)	(1 mark)